

Communication-Constrained Soft-Information Fusion for Multi-UAV OTFS-ISAC Cooperative Sensing

Xinhang Tan, Zhengquan Zhang, L Hao
Southwest Jiaotong University
Chengdu, China

Email: tanxinhang6478@163.com, zhang.zhengquan@swjtu.edu.cn, lhao@home.swjtu.edu.cn

Abstract—In multi-UAV OTFS-ISAC networks, inter-UAV soft-information exchange can improve cooperative sensing, but excessive signaling overhead, exchange delay, and unreliable reporting links may offset the sensing gain. To address this issue, this paper proposes a communication-error-calibrated and coarse-to-fine (C2F) delay-Doppler (DD)-aware Lagrangian link-selection method for communication-constrained multi-UAV cooperative sensing. The proposed method evaluates each target-dependent cooperative link according to its calibrated marginal deflection gain, reporting reliability, exchange delay, and DD-domain energy-capture efficiency. A coarse DD-domain Lagrangian utility is first used to screen candidate links, and local DD-grid refinement is then applied only to shortlisted candidates, thereby reducing unnecessary fine-grid evaluations. Simulation results in a 15-UAV, 10-target aerial sensing scenario show that the proposed method retains about 97.2% of the all-neighbor detection probability while reducing signaling overhead and exchange delay by approximately 96.7% and 99.0%, respectively. Under the same per-target selected-link budget, it outperforms Sensing-SINR-based selection and achieves comparable detection to Top-K Deflection with about 6.5% lower exchange delay. Because the communication and sensing receivers share one interference field, the sensing task requires direct-path suppression of the order of the near-far ratio (40 dB here); below that the detection probability collapses irrespective of how the links are selected.

Index Terms—Multi-UAV networks, OTFS-ISAC, cooperative sensing, soft-information fusion, DD-domain refinement, Lagrangian link selection.

I. INTRODUCTION

Integrated sensing and communication (ISAC) enables wireless networks to reuse spectrum, hardware, and signal-processing resources for both data transmission and environmental sensing [1], [2]. UAV-enabled ISAC extends this capability to aerial sensing scenarios, where platform mobility can improve coverage, line-of-sight availability, and sensing geometry [5], [6]. When multiple UAVs cooperate, the same target can be observed from different bistatic geometries, providing complementary spatial information for more reliable sensing decisions [3].

In such a distributed network, however, inter-UAV reporting links have limited rate and reliability, so more cooperative observations do not necessarily lead to higher fusion gain [3],

[9]. Each observation is generated by a specific transmitter–receiver–target geometry, reported over an inter-UAV link, and then fused with the others, and its contribution can be weakened at any stage of this sensing–reporting–fusion process. A geometrically favorable return may suffer from inefficient delay-Doppler (DD)-domain energy capture [10], [11]; its soft statistic may be distorted by unreliable reporting [7], [8]; or its reporting cost may be too large relative to its marginal fusion gain [4], [9]. As a result, all-neighbor fusion or sensing-SINR-based activation can be inefficient when inter-UAV communication resources are limited.

Existing studies have examined these aspects, though largely from separate viewpoints. Aerial sensing architectures, mobility, and resource management are the main focus of UAV-ISAC work [5], [6]. Sensing–communication tradeoffs under resource or backhaul constraints, by contrast, have been considered in network-level ISAC and cooperative sensing studies [4], [9]. On the detection side, imperfect channel knowledge and unreliable delivery have been shown to degrade detection performance [7], [8]. Separately, OTFS-based ISAC exploits DD-domain structures for high-mobility sensing and communication [10], [11]. Each line captures one facet of the problem, yet communication-constrained triplet-level scheduling that couples these facets for multi-UAV bistatic OTFS-ISAC has received limited attention.

This gap motivates a triplet-level scheduling perspective. Node-selection and resource-allocation studies act at the level of UAV nodes or links [9], and distributed-detection studies model reporting reliability without the underlying bistatic geometry [7], [8]; neither, however, exposes the unit that actually carries a cooperative observation. In multi-UAV bistatic sensing that unit is a target-dependent transmitter–receiver–target triplet, whose value depends jointly on its bistatic sensing contribution, the reliability and delay of its reporting link, and the DD-domain energy-capture efficiency of the corresponding OTFS observation [10]. Judging these factors in isolation is misleading—a triplet with high nominal sensing-SINR can still add little after fusion if its soft statistic is reported over a slow or unreliable channel, or if its DD-domain energy is poorly captured. We therefore treat each triplet as a schedulable soft-

information reporting unit and score it on a calibrated sensing contribution, reporting reliability, exchange delay, and DD-domain efficiency taken jointly, aiming to select a compact set of cooperative observations that preserves most of the sensing gain while reducing signaling overhead and reporting delay.

Our main contributions are as follows.

- We formulate communication-constrained multi-UAV OTFS-ISAC cooperative sensing as a triplet-level soft-information link-selection problem. Each candidate is characterized by its bistatic sensing contribution, reporting reliability, exchange delay, and DD-domain energy-capture efficiency.
- We develop a communication-error-calibrated marginal-deflection metric and a C2F DD-aware Lagrangian link-selection method. The method evaluates target-dependent candidates by balancing marginal fusion gain and reporting delay, while applying local DD-grid refinement only to shortlisted links.
- Simulation results in a 15-UAV, 10-target aerial sensing scenario show that the proposed method retains most of the all-neighbor detection gain with substantially lower signaling overhead and reporting delay. It also provides a more favorable detection–delay tradeoff than sensing-SINR-based selection.

The rest of this paper is organized as follows. Section II presents the multi-UAV OTFS-ISAC system model. Section III introduces the proposed C2F DD-aware Lagrangian link-selection method. Section IV provides simulation results and discussion. Section V concludes the paper.

II. SYSTEM MODEL

We consider a sensing-centric multi-UAV OTFS-ISAC network, as shown in Fig. 1(a). The UAV and target sets are denoted by $\mathcal{M} = \{1, \dots, M\}$ and $\mathcal{Q} = \{1, \dots, Q\}$, respectively. UAV i and target q have position–velocity pairs $(\mathbf{p}_i, \mathbf{v}_i)$ and $(\mathbf{p}_q, \mathbf{v}_q)$. A cooperative candidate is indexed by the target-dependent triplet (i, j, q) , where UAV i transmits an OTFS-ISAC probing signal, target q reflects it, and UAV j receives the bistatic echo. The inter-UAV reporting topology is represented by a directed graph $\mathcal{G} = (\mathcal{M}, \mathcal{E})$, where $(i, j) \in \mathcal{E}$ if $\|\mathbf{p}_i - \mathbf{p}_j\| \leq d_{\max}$ and $i \neq j$.

We explicitly separate the *sensing* topology from the *reporting* topology. The sensing chain is $i \rightarrow q \rightarrow j$: UAV j receives the echo and therefore *produces* the soft statistic. The statistic is then reported over the link $j \rightarrow f_q$ to a target-specific fusion UAV $f_q \in \mathcal{M}$, which is the single node where all soft statistics about target q meet. Accordingly, a sensing-reporting candidate is the quadruple $(i, j, q; f_q)$, and the communication quantities that matter—reliability $\chi_{j f_q}$, rate $R_{j f_q}$ and delay $T_{j f_q}$ —are properties of the reporting leg (j, f_q) alone, with no dependence on the illuminator i . This decoupling is what guarantees the fused statistic is physically well defined: with a naive $j \rightarrow i$ reporting rule, three selected links (i_1, j_1, q) , (i_2, j_2, q) , (i_3, j_3, q) would deliver their statistics to three different UAVs while the fusion step implicitly assumed they had already met. We use a geometry-driven soft-information

abstraction rather than a full waveform-level OTFS receiver model. The OTFS structure enters the scheduling problem through DD-bin validity, fractional DD energy loss, same-bin collision, and local DD-grid refinement. These DD-domain quantities determine whether a target-dependent triplet provides useful soft information after reporting and fusion.

A. Communication and Reporting Model

The transmit power of UAV i is split into sensing and communication parts, i.e.,

$$P_i^s = \rho P_i, \quad P_i^c = (1 - \rho) P_i. \quad (1)$$

For a directed inter-UAV link (a, b) , the communication-channel power gain is

$$g_{ab}^c = \left(\frac{\lambda}{4\pi} \right)^2 d_{ab}^{-\alpha_c} \zeta_{ab} |h_{ab}^c|^2, \quad d_{ab} = \|\mathbf{p}_a - \mathbf{p}_b\|. \quad (2)$$

Here λ , α_c , ζ_{ab} , and h_{ab}^c denote the carrier wavelength, path-loss exponent, shadowing, and small-scale fading, respectively.

The communication SINR from UAV i to UAV j is modeled as

$$\gamma_{ij}^c = \frac{P_i^c g_{ij}^c}{N_0 + \sum_{k \in \mathcal{M} \setminus \{i, j\}} (P_k^c + \epsilon_s P_k^s) g_{kj}^c + \epsilon_d P_i^s g_{ij}^c}. \quad (3)$$

The coefficients ϵ_s and ϵ_d represent neighbor-induced sensing-to-communication leakage and intra-link leakage from the sensing component of UAV i to the communication receiver, respectively.

One interference field, two receivers. Because each UAV radiates a *single* joint waveform into a shared band, the very same concurrent transmitters reach the communication and the sensing receiver through the very same channel (2). Let $\mathcal{A} \subseteq \mathcal{M}$ be the set of UAVs that actually transmit a report payload; the radiated ISAC power of UAV k is then

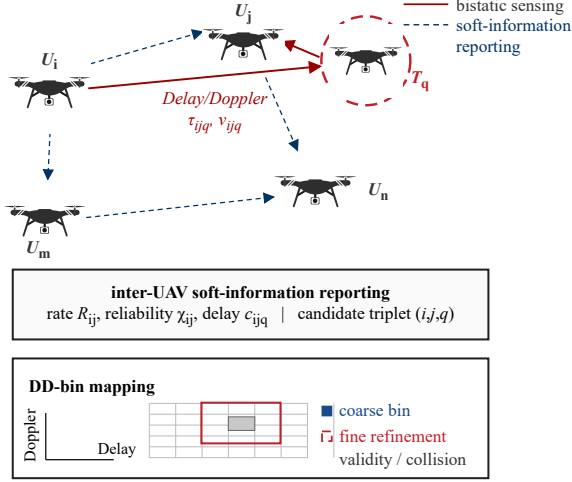
$$P_k^{\text{tx}} = P_k^s + \mathbb{I}\{k \in \mathcal{A}\} P_k^c, \quad (4)$$

since the sensing waveform is radiated continuously, and the two receivers at node j observe

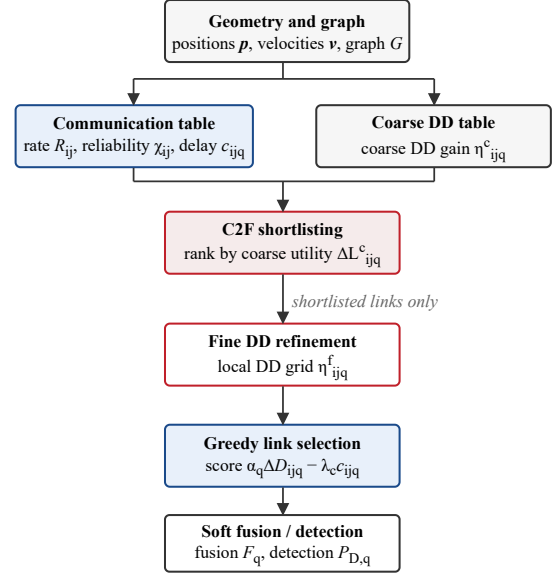
$$\begin{aligned} I_j^c &= \sum_{k \in \mathcal{M} \setminus \{j\}} (\mathbb{I}\{k \in \mathcal{A}\} P_k^c + \epsilon_s P_k^s) g_{kj}^c, \\ I_j^s &= \kappa_{\text{self}} P_j^s + \kappa_{\text{dc}} \sum_{k \in \mathcal{M} \setminus \{j\}} P_k^{\text{tx}} g_{kj}^c. \end{aligned} \quad (5)$$

Giving the communication receiver a full interference term while treating the sensing interference as an unstructured floor is not an ISAC model: it removes precisely the coupling that makes the two functions worth co-designing. Setting $\mathcal{A} = \mathcal{M}$ recovers the full-concurrent form (3); the reported results use the scheduled form, in which only UAVs that actually report contribute their payload, whereas the sensing waveform—being what produces the echoes in the first place—is always radiated. Two structural properties are worth stating. First, the illuminator i is an interferer on the sensing side, because its direct path competes with its own target echo; on the communication side the same UAV is the wanted source.

(a) Multi-UAV OTFS-ISAC Scenario



(b) C2F DD-Aware Selection and Fusion



Coarse table: screening; fine table: final selection, fusion, and detection.

Fig. 1. Multi-UAV OTFS-ISAC cooperative sensing scenario and C2F DD-aware link-selection framework.

This is a difference in the role a node plays, not in the physics. Second, κ_{dc} is not a tuning floor but the *direct-path cancellation* the sensing receiver achieves: cooperative ISAC knows every illuminator's waveform, so the deterministic direct path can be reconstructed and subtracted, and only the channel-estimation residual remains.

The suppression κ_{dc} cannot be chosen freely, because the interference and the useful signal do not scale alike. By (16) the echo suffers a fourth-power path loss while the direct path suffers a second-power one, so their ratio

$$\rho_{NF} = \frac{\sum_{k \in \mathcal{M} \setminus \{j\}} P_k^{\text{tx}} g_{kj}^c}{\tilde{P}_i^s g_{ijq}^s G_p} \quad (6)$$

grows as d^2 and is $\rho_{NF} \approx 40$ dB for the geometry of Section IV. The sensing task therefore survives only if $\kappa_{dc} \lesssim \rho_{NF}^{-1}$: the required suppression budget is itself the price that sharing the spectrum with the communication function imposes on the sensing receiver. We accordingly treat κ_{dc} as a first-class system parameter and sweep it, rather than asserting a value.

With communication bandwidth $B_c = N\Delta f$, the reporting rate and exchange delay are

$$R_{ij} = B_c \log_2(1 + \gamma_{ij}^c), \quad c_{ijq} = \frac{B_q}{R_{ij}}, \quad (7)$$

where $B_q = K_{\text{pkt}} b_d$ is the soft-information payload size. A reporting link is feasible when $R_{ij} \geq R_{\min}$.

Finite-blocklength reliability. A soft sensing report is a short control packet. The default simulation uses the compact abstraction

$$\chi_{ij}^{\text{heur}} = \frac{\gamma_{ij}^c}{\gamma_{ij}^c + \gamma_{\text{req}}}, \quad \gamma_{\text{req}} = 2^{R_{\min}/B_c} - 1, \quad (8)$$

which is a smooth monotone map but carries no probabilistic meaning. Its strictly probabilistic counterpart is the finite-blocklength packet success probability: the packet error probability is, by the normal approximation,

$$\epsilon_{ij} \approx Q \left(\frac{C(\gamma_{ij}^c) - \frac{k}{n}}{\sqrt{V(\gamma_{ij}^c)/n}} \right), \quad \chi_{ij} = 1 - \epsilon_{ij}, \quad (9)$$

where $C(\gamma) = \log_2(1 + \gamma)$, $V(\gamma) = (1 - (1 + \gamma)^{-2}) (\log_2 e)^2$, $k = B_q$ is the payload in bits and n is the blocklength in channel uses. The report therefore has a genuine *packet success probability* χ_{ij} , and reliability, payload and latency become one coupled system: k enters the error exponent, while n enters both the error exponent and the block latency $T_{ij} = n/B_c$.

B. DD-Domain Bistatic Sensing Model

For triplet (i, j, q) , define $d_{iq} = \|\mathbf{p}_q - \mathbf{p}_i\|$, $d_{jq} = \|\mathbf{p}_q - \mathbf{p}_j\|$, $\mathbf{u}_{iq} = (\mathbf{p}_q - \mathbf{p}_i)/d_{iq}$, and $\mathbf{u}_{jq} = (\mathbf{p}_q - \mathbf{p}_j)/d_{jq}$. The bistatic delay and Doppler shift are

$$\tau_{ijq} = \frac{d_{iq} + d_{jq}}{c}, \quad \nu_{ijq} = \frac{(\mathbf{v}_q - \mathbf{v}_i)^T \mathbf{u}_{iq} + (\mathbf{v}_q - \mathbf{v}_j)^T \mathbf{u}_{jq}}{\lambda}. \quad (10)$$

The corresponding DD-grid indices are

$$l_{ijq} = \text{round}(\tau_{ijq}L\Delta f), \quad k_{ijq} = \text{round}(\nu_{ijq}NT). \quad (11)$$

A triplet is DD-valid if $0 \leq l_{ijq} < L$ and $-N/2 \leq k_{ijq} < N/2$. The fractional DD offsets are

$$\varepsilon_{ijq}^\tau = \tau_{ijq}L\Delta f - l_{ijq}, \quad \varepsilon_{ijq}^\nu = \nu_{ijq}NT - k_{ijq}. \quad (12)$$

Using the normalized sinc function, the coarse DD-domain gain is

$$\eta_{ijq}^c = \text{sinc}^2(\varepsilon_{ijq}^\tau) \text{sinc}^2(\varepsilon_{ijq}^\nu). \quad (13)$$

For C2F DD refinement, a local $(2W+1) \times (2W+1)$ DD neighborhood provides the recoverable energy η_{ijq}^{loc} , and the refined gain is

$$\eta_{ijq}^f = \min\{1, \max[\eta_{\min}, \eta_{ijq}^c + \kappa_{\text{dd}}(\eta_{ijq}^{\text{loc}} - \eta_{ijq}^c)]\}. \quad (14)$$

Here $\kappa_{\text{dd}} \in [0, 1]$ controls the refinement strength and $\eta_{\min}$ is the retained-energy floor.

If multiple DD-valid targets observed by the same UAV pair fall into the same DD bin, we use the collision penalty

$$\eta_{ijq}^{\text{col}} = \frac{1}{n_{ijq}^{\alpha_{\text{col}}}}, \quad (15)$$

where n_{ijq} is the number of targets sharing bin (l_{ijq}, k_{ijq}) .

The bistatic target gain is

$$g_{ijq}^s = \frac{\lambda^2 \sigma_q}{(4\pi)^3 d_{iq}^2 d_{jq}^2}, \quad (16)$$

where σ_q denotes the target RCS. Combining bistatic gain, processing gain, DD-domain gain, and collision penalty gives

$$\gamma_{ijq}^s = \frac{\tilde{P}_i^s g_{ijq}^s G_p \eta_{ijq}^{\text{DD}} \eta_{ijq}^{\text{col}}}{N_0 + I_j^s}, \quad \eta_{ijq}^{\text{DD}} \in \{\eta_{ijq}^c, \eta_{ijq}^f\}. \quad (17)$$

In the default sensing-only mode, $\tilde{P}_i^s = P_i^s$, and $G_p = NL$. The interference term is the *same field* (5) that the communication receiver at j sees, attenuated only by the direct-path cancellation κ_{dc} that the sensing receiver achieves; the illuminator's own direct path is part of it, which is what makes (17) a genuine near-far limited problem rather than an interference-free one.

C. Soft-Information Fusion Model

Each feasible triplet produces a scalar soft statistic s_{ijq} . The default simulation uses the compact Gaussian abstraction

$$\mu_{ijq} = \kappa_\mu \log(1 + \gamma_{ijq}^s), \quad \sigma_{ij,0}^2 = \sigma_0^2 (1 + \xi I_j^s / N_0), \quad (18)$$

with a hand-set scale κ_μ ; this encodes the tautology ‘‘a higher sensing SINR gives a larger soft mean’’ and nothing more.

Strictly derived local LLR. Because κ_μ carries no physical meaning, we also provide the strictly derived statistic. After OTFS matched filtering, the content of the delay-Doppler bin carrying target q on link $i \rightarrow q \rightarrow j$ is a single complex sample with a Swerling-I fluctuation. The normalised bin energy $x = |z|^2 / \sigma_n^2$ is therefore exponential, $x \sim \text{Exp}(1)$ under $\mathcal{H}_0$ and

$x \sim \text{Exp}(1 + \gamma_{ijq}^s)$ under $\mathcal{H}_1$. The local log-likelihood ratio, centred under $\mathcal{H}_0$, is

$$\ell_{ijq} = \frac{\gamma_{ijq}^s}{1 + \gamma_{ijq}^s} (x - L), \quad (19)$$

where L is the number of OTFS frames incoherently integrated in one CPI. Its moments are

$$\begin{aligned} \delta_{ijq} &= \mathbb{E}_1[\ell] - \mathbb{E}_0[\ell] = \frac{L(\gamma_{ijq}^s)^2}{1 + \gamma_{ijq}^s}, \\ \sigma_{ij,0}^2 &= \text{Var}_0[\ell] = \frac{L(\gamma_{ijq}^s)^2}{(1 + \gamma_{ijq}^s)^2}, \end{aligned} \quad (20)$$

and the per-link information gain is a Kullback-Leibler divergence

$$J_{ijq} = D_{\text{KL}}(p_1 \| p_0) = L(\gamma_{ijq}^s - \ln(1 + \gamma_{ijq}^s)). \quad (21)$$

This strictly derived statistic carries no free scale parameter, and its single-link deflection $\delta_{ijq}^2 / \sigma_{ij,0}^2 = L(\gamma_{ijq}^s)^2$ is Swerling's classical result. The Gaussian abstraction (18) is retained as the default because it isolates the scheduling behaviour from the detector-level operating point; the LLR model is validated in Section IV.

Communication-error calibration acts on the soft statistic through the reporting leg $j \rightarrow f_q$:

$$\delta_{ijq}^{\text{eff}} = \chi_{jf_q} \delta_{ijq}, \quad \bar{\sigma}_{ij,0}^2 = \chi_{jf_q} \sigma_{ij,0}^2 + (1 - \chi_{jf_q}) \sigma_{\text{err}}^2, \quad (22)$$

where χ_{jf_q} is the success probability (9) of the report $j \rightarrow f_q$.

For selected set $\mathcal{S}_q$, let $\delta_q = (\delta_{ijq}^{\text{eff}})_{(i,j) \in \mathcal{S}_q}$ and let Σ_q be the covariance of the stacked local statistics. The default assumes the local statistics are independent, giving

$$D_q(\mathcal{S}_q) = \sum_{(i,j) \in \mathcal{S}_q} \frac{(\delta_{ijq}^{\text{eff}})^2}{\bar{\sigma}_{ij,0}^2}. \quad (23)$$

In practice the local statistics are correlated (shared transmitters, shared receivers, the shared Swerling fluctuation of target q and possibly neighbouring DD support). The correlation-aware fusion generalises to

$$D_q(\mathcal{S}_q) = \delta_q^T \Sigma_q^{-1} \delta_q, \quad \mathbf{w}_q^* \propto \Sigma_q^{-1} \delta_q, \quad (24)$$

which collapses to (23) when Σ_q is diagonal. Correlated, redundant UAV observations are thus down-weighted by the mathematics rather than by a hand-set diversity factor.

Approximating F_q as Gaussian, the detection probability is

$$P_{D,q} \approx Q\left(Q^{-1}(P_{\text{FA}}) - \sqrt{D_q(\mathcal{S}_q)}\right), \quad (25)$$

where $Q(\cdot)$ denotes the Gaussian Q -function.

D. Truth, Belief and the Scheduling Loop

Detection has not happened when the scheduler selects links, so the scheduler cannot be given the true target state. We therefore carry two states side by side. The *truth* $(\mathbf{p}_{q,t}, \mathbf{v}_{q,t})$ drives echo generation, the true delay-Doppler, the true sensing gain and the detector; the *belief* $b_{q,t} = \mathcal{N}(\hat{\mathbf{x}}_{q,t}, \mathbf{P}_{q,t})$ drives

link selection, the DD search window and the resource budget. The loop is

$$\begin{aligned} & \text{predict} \rightarrow \text{schedule} \rightarrow \text{sense} \rightarrow \text{update}, \\ & \hat{\mathbf{x}}_{t|t-1} = \mathbf{F}\hat{\mathbf{x}}_{t-1|t-1}, \\ & \mathbf{P}_{t|t-1} = \mathbf{F}\mathbf{P}_{t-1|t-1}\mathbf{F}^T + \mathbf{Q}. \end{aligned} \quad (26)$$

A selected link delivers target evidence only when the belief-guided search window captures the true delay-Doppler bin; a mismatched belief therefore degrades detection exactly as a reviewer would expect, and this is quantified in Section IV through a belief-mismatch sweep.

III. PROPOSED C2F DD-AWARE LAGRANGIAN LINK SELECTION

We select a compact set of cooperative candidates whose marginal fusion gain justifies their reporting cost. The selection is implemented as a low-complexity greedy rule guided by a Lagrangian marginal-gain score. For target q , the feasible candidate set is

$$\mathcal{E}_q = \{(i, j) \in \mathcal{E} : R_{ij} \geq R_{\min}, (l_{ijq}, k_{ijq}) \text{ is DD-valid}\}. \quad (27)$$

Here c_{ijq} and the DD-valid condition are defined in Section II. We use $\tilde{c}_{ijq} = 10^3 c_{ijq}$ as the delay in milliseconds.

Given the current selected set $\mathcal{S}_q$, the target-priority coefficient is updated as

$$\alpha_q = \frac{\partial P_{D,q}}{\partial D_q} + \mu_\alpha \frac{[D_{\min} - D_q(\mathcal{S}_q)]^+}{D_{\min}}, \quad (28)$$

where $[x]^+ = \max(x, 0)$. The derivative follows from (25) with a small regularization when D_q is close to zero, and $\alpha_q^{(0)}$ denotes the value evaluated at $\mathcal{S}_q = \emptyset$.

For candidate (i, j) and target q , the marginal calibrated deflection gain is

$$\Delta D_{ijq} = D_q(\mathcal{S}_q \cup \{(i, j)\}) - D_q(\mathcal{S}_q), \quad (29)$$

where $D_q(\cdot)$ uses the calibrated statistics in (22). The Lagrangian score is

$$\Delta \mathcal{L}_{ijq} = \alpha_q \Delta D_{ijq} - \lambda_c \tilde{c}_{ijq}, \quad (30)$$

where $\lambda_c > 0$ controls the delay penalty.

To reduce local DD-grid refinement cost, we use a C2F strategy. Let L_{all} be the number of feasible candidates before screening and $L_{\text{short}} \ll L_{\text{all}}$ be the shortlist size. In the coarse stage, candidates are ranked by

$$\Delta \mathcal{L}_{ijq}^c = \alpha_q^{(0)} \Delta D_{ijq}^c - \lambda_c \tilde{c}_{ijq}, \quad (31)$$

where ΔD_{ijq}^c is computed using the coarse DD gain in (13). Only the top L_{short} candidates are retained for fine refinement.

The fine stage applies (14) only to the shortlisted candidates and recomputes (29) and (30). Starting from $\mathcal{S}_q = \emptyset$, the algorithm repeatedly selects the remaining candidate with the largest positive $\Delta \mathcal{L}_{ijq}$ and updates $D_q(\mathcal{S}_q)$ and α_q . It stops when no positive-score candidate exists, the link-number limit is reached, or all active targets with feasible candidates satisfy $D_q(\mathcal{S}_q) \geq D_{\min}$. C2F reduces fine DD-grid evaluations from $\mathcal{O}(QL_{\text{all}}(2W+1)^2)$ to $\mathcal{O}(QL_{\text{short}}(2W+1)^2)$.

A. Optimality Guarantee of the Greedy Rule

The greedy rule (30) is precisely the first-order greedy step on the scalar objective

$$F(\mathcal{S}) = \sum_{q \in \mathcal{Q}} P_{D,q}(D_q(\mathcal{S}_q)) - \lambda_c \sum_{(i,j) \in \mathcal{S}} \tilde{c}_{ijq}, \quad (32)$$

since $\alpha_q = \partial P_{D,q} / \partial D_q$ is the local slope of the detection utility in (28). The structure of F justifies the greedy rule. When the local statistics are independent, the deflection (24) is additive, $D_q = \sum_{(i,j)} \delta_{ijq}^2 / \bar{\sigma}_{ij,0}^2$, so $P_{D,q}(D_q)$ is a monotone concave function of a modular function and is therefore submodular. With the correlation model enabled, $D_q = \boldsymbol{\delta}_q^T \boldsymbol{\Sigma}_q^{-1} \boldsymbol{\delta}_q$ is the conditional information gain of an exponential-family model and is likewise submodular in the link set. Since the reporting cost is linear in $\mathcal{S}$, F is a submodular function minus a linear cost, and the cost-agnostic greedy selection attains the standard curvature guarantee

$$F(\mathcal{S}_{\text{greedy}}) \geq \frac{1 - e^{-c}}{c} F(\mathcal{S}^*), \quad (33)$$

where $c \in [0, 1]$ is the total curvature of the utility and $\mathcal{S}^*$ is the optimal set under the link budgets. The factor interpolates between $1 - 1/e$ at $c = 1$ and exact optimality at $c = 0$ (the modular case). Section IV verifies the diminishing-returns property numerically and reports the empirical curvature and the implied guarantee, together with the small-scale greedy-vs-oracle gap on the *same* objective (32).

IV. SIMULATION RESULTS

We evaluate the proposed communication-error-calibrated and C2F DD-aware Lagrangian link-selection scheme in a high-mobility multi-UAV aerial sensing scenario. Unless otherwise specified, 15 UAVs cooperatively sense 10 aerial targets in a $4000 \text{ m} \times 4000 \text{ m}$ area. The UAVs and targets move with speeds of 30–60 m/s and 50–90 m/s, respectively. We use the all-neighbor scheme as an upper-resource reference and set the target false-alarm probability to $P_{\text{FA}} = 0.05$ in all tests. Table I lists the main simulation parameters.

A. Performance and Tradeoff Analysis

For the main comparison in Fig. 2, the proposed method achieves $P_D = 0.9481$, which retains about 97.2% of the all-neighbor detection probability. At the same time, it reduces signaling overhead and exchange delay by approximately 96.7% and 99.0%, respectively. In absolute terms, the exchange delay decreases from about 6.9×10^2 ms under all-neighbor fusion to about 6.9 ms with the proposed method. This confirms that all-neighbor fusion is a useful upper-resource reference, but its reporting cost is too high for low-latency cooperative sensing.

A major source of this gain is the avoidance of ineffective soft-information reporting. Sensing-SINR-based selection tends to choose links with strong sensing responses, without explicitly checking whether their soft statistics can be reported reliably and quickly. By combining communication-error-calibrated marginal deflection with delay-aware Lagrangian

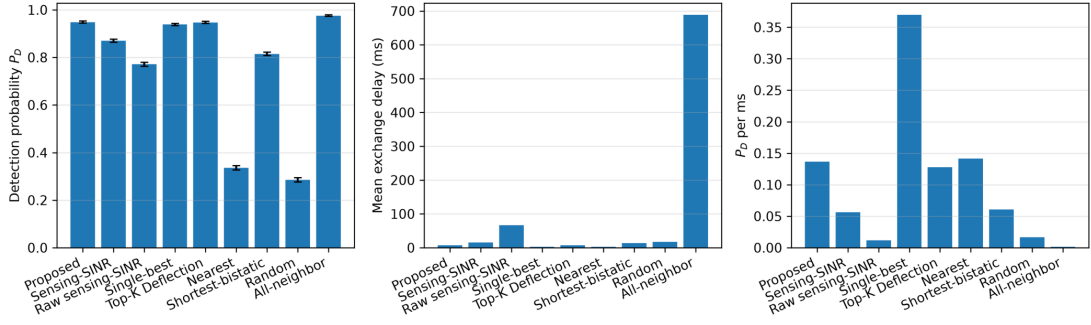


Fig. 2. Main performance comparison under C2F DD refinement.

TABLE I
MAIN SIMULATION PARAMETERS

Parameter	Value
UAVs / targets	$M = 15, Q = 10$
Area and comm. range	$4000 \times 4000 \text{ m}^2, d_{\max} = 2500 \text{ m}$
UAV / target speed	$30\text{--}60 \text{ m/s} / 50\text{--}90 \text{ m/s}$
Carrier and DD grid	$f_c = 5.9 \text{ GHz}, N = L = 64, \Delta f = 30 \text{ kHz}$
Power split / Tx power	$\rho = 0.8, P_i = 1 \text{ W}$
Comm. channel	$\alpha_c = 2$, shadowing std. 1 dB, Rician $K = 15 \text{ dB}$
Rate constraint / false alarm	$R_{\min} = 0.2 \text{ Mbit/s}, P_{FA} = 0.05$
Payload size	$K_{\text{pkt}} = 4, b_d = 160 \text{ bits}, B_q = 640 \text{ bits}$
Link-selection parameters	$D_{\min} = 3, \lambda_c = 0.005, L_{\max} = 6, L_{\text{tot}} = 60$
ISAC interference coupling	shared-spectrum field, $\kappa_{dc} = 40 \text{ dB}$, guard $N_0 \cdot 10^{-3}$ (−30 dB rel. N_0)
C2F DD parameters	$W = 1, \kappa_{dd} = 0.45, \eta_{\min} = 0.25, L_{\text{short}} = 25$
Monte Carlo trials	1000

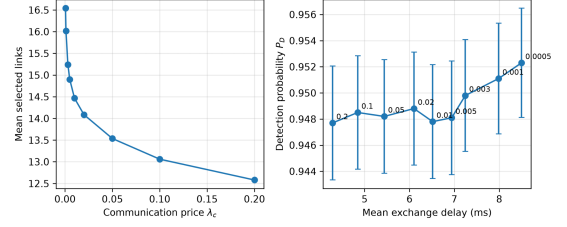


Fig. 3. Sensitivity to the Lagrangian communication price λ_c .

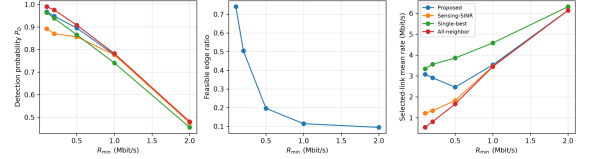


Fig. 4. Sensitivity to the communication-rate constraint $R_{\min}$.

pricing, the proposed method keeps most useful cooperative observations and discards links whose reporting cost is not justified by their effective fusion contribution; it attains 0.9481 against 0.8705 for Sensing-SINR and 0.8148 for Shortest-Bistatic, a margin of 7.8 percentage points over the strongest competing rule.

Under the same per-target selected-link budget, the proposed method achieves higher detection probability than every baseline except the all-neighbor upper reference, and it marginally exceeds Top-K Deflection (0.9481 vs. 0.9470) while reducing exchange delay by about 6.5%. This behavior is consistent with the delay-aware Lagrangian term, which discourages unreliable or slow soft-information links even when their sensing or deflection scores are high.

Varying λ_c in Fig. 3 changes the selected-link pattern directly. Over more than two decades of λ_c (from 5×10^{-4} to 0.2) the number of selected links falls from 16.5 to 12.6, the exchange delay from 8.5 to 4.3 ms and the overhead from 10.6 to 8.1 kbit, while P_D changes by less than half a percentage point (0.9523 to 0.9477). Hence λ_c is a practical control knob for the detection–delay tradeoff that buys latency and overhead almost for free. Changing $R_{\min}$ in Fig. 4 affects the

problem in a different way: a larger rate requirement shrinks the feasible reporting graph and weakens the cooperative sensing gain. P_D falls from 0.9678 at $R_{\min} = 0.1$ to 0.4797 at $R_{\min} = 2$ (in Mbit/s). In the low-to-moderate constraint region the proposed method still maintains a better sensing–communication tradeoff than Sensing-SINR by prioritizing reliable and low-delay links: at $R_{\min} = 0.1$ it attains 0.9678 against 0.8927, and it only converges to the baseline once the feasible graph is so sparse that almost no selection freedom remains.

Taken together, these two sensitivity tests separate the roles of algorithmic pricing and communication feasibility. The Lagrangian price adjusts the penalty on reporting delay, whereas the rate constraint changes which inter-UAV links can be used for soft-information exchange. When the feasible reporting graph becomes too sparse, all cooperative schemes degrade, indicating that communication feasibility remains a fundamental bottleneck. We note that under the corrected interference model the fraction of feasible reporting edges at the nominal $R_{\min} = 0.2$ rises from 0.33 to 0.50, because the scale-relative SINR guard restores the communication SINR on weakly interfered links. The nominal operating point is therefore only moderately constrained; the constraint becomes

TABLE II
COMPONENT ABLATION UNDER THE SAME SELECTED-LINK BUDGET

Variant	P_D	Bits (kbit)	Delay (ms)	Worst P_D
Full	0.9469	9.17	6.56	0.938
w/o target priority	0.8504	15.88	9.40	0.826
w/o delay price	0.9482	9.43	7.18	0.938
w/o comm.-error calib.	0.8445	6.57	8.15	0.830
w/o softmin- α	0.9540	10.00	7.18	0.947

binding for $R_{\min} \gtrsim 1$, where P_D falls to 0.78 and the selection freedom effectively disappears.

B. Ablation and Robustness Analysis

For the component study in Table II, all variants are evaluated under a fixed selected-link budget. The “Full” row is therefore used for within-table comparison and is not intended to duplicate the main-comparison value in Fig. 2. Removing communication-error calibration causes the largest degradation, reducing P_D from 0.9469 to 0.8445, which highlights the importance of calibrating unreliable exchanged soft information before fusion. Removing target priority is comparably damaging (0.8504) and additionally inflates the overhead from 9.17 to 15.88 kbit and the delay from 6.56 to 9.40 ms, confirming that the priority term is what keeps the selector from over-investing in targets that are already resolved. Without delay-aware pricing P_D is marginally higher (0.9482), but the exchange delay grows by about 9%; this matches the role of the delay-price term in trading a small amount of detection gain for lower reporting latency. Notably, removing the softmin weighting raises P_D from 0.9469 to 0.9540 but costs more overhead and delay (10.00 kbit and 7.18 ms), indicating that the softmin- α fairness term currently lacks experimental support; it is retained only as a configurable mechanism and its justifying functional is left to future work.

In a separate DD-domain ablation under the main-comparison configuration, C2F refinement improves P_D from 0.9481 to 0.9572. It matches full local refinement (0.9537, the difference being within Monte-Carlo error) while reducing fine DD-grid evaluations by about 44.6%. This experiment isolates the effect of DD-domain refinement and shows that local DD-grid recovery is useful, whereas applying it to every feasible triplet is unnecessary.

Robustness is evaluated along four axes in Fig. 5. Under communication errors the proposed method attains $P_D = 0.9481$ for erasure, 0.9756 for the biased model and 0.9050 for bit flipping, so even the worst case retains 92.8% of the error-free value. Sweeping the soft-error scale from 1 to 4 degrades P_D only from 0.9488 to 0.9446, at the cost of growing the exchange delay from 4.9 to 8.3 ms. The two interference axes are qualitatively different: the residual-floor axis is evaluated inside the decoupled model, where it controls the sensing interference directly and drives P_D from 0.6550 down to 0.4970 over two decades, whereas the direct-path cancellation axis is the ISAC-specific axis analysed in Section IV-D. In every case the proposed method maintains a

favorable detection–delay tradeoff because communication reliability and interference-aware calibration penalize unreliable links before fusion. These results support the robustness of the proposed soft-information fusion strategy under imperfect reporting conditions.

C. Model Refinements and Theoretical Guarantees

This subsection validates the stricter model components introduced in Section II and the structural guarantee of the greedy rule.

Strictly derived soft statistic (local LLR). Replacing the Gaussian abstraction (18) by the waveform-derived local LLR (19)–(21) removes the hand-set scale κ_μ . Under the LLR model the proposed method attains $P_D = 0.9048$ with all-neighbor $P_D = 0.9513$, i.e. a 95.1% retention of the upper reference, and it still outperforms the Sensing-SINR (0.8360) and Top-K Deflection (0.9034) baselines while lowering the exchange delay from 19.2 to 9.9 ms. The absolute level is 4.3 points below the Gaussian abstraction, which is the price of deleting the $\kappa_\mu = 8$ scale; the ordering of the methods is preserved, confirming that the scheduling behaviour is not an artefact of that scale, while the per-link information gain is exactly the KLD (21).

Finite-blocklength reliability. With the FBL model (9), the mean report reliability grows from $\chi \approx 0.404$ at blocklength $n = 256$ to $\chi \approx 0.918$ at $n = 512$ and $\chi \approx 1.0$ from $n = 1024$ onwards, while the block latency n/B_c grows from 0.67 ms to 22.3 ms. Correspondingly, P_D rises from 0.140 to 0.512, 0.865, 0.960 and 0.987 at $n = 4096$. The FBL sweep thus exposes the reliability–latency trade-off that the heuristic map (8) cannot express: reliability, payload and latency are one coupled system, and the useful operating range is $n \in [1024, 4096]$, where reliability has saturated but latency is still affordable.

Correlation-aware fusion. Enabling the observation-covariance fusion (24) is essentially neutral under the corrected model: P_D moves from 0.9481 to 0.9448, the mean deflection from 25.10 to 24.49, and the selected-link count from 14.9 to 15.7. This is consistent with the corrected selector already working close to the minimum link count, so little redundant observation is left for the covariance model to prune; the main value of the model is that it makes the assumption explicit rather than silently double-counting correlated pairs.

Truth vs. belief. Under the belief-mismatch sweep (26), detection degrades monotonically as the tracker belief error grows: $P_D = 0.9481$ at a perfect tracker falls to 0.3784 at belief error $\sigma_p = 50$ m, to 0.0762 at 150 m and to 0.0249 at 300 m, with the belief-window capture rate dropping from 1.00 to 0.37, 0.07 and 0.02. This quantifies the predict–schedule–sense–update cost that the earlier “perturbed-world” experiments could not capture, and it also identifies the tracking accuracy as the binding constraint once the reporting budget is resolved.

Submodularity and the greedy guarantee. The diminishing-returns audit on 800 sampled set pairs reports zero mono-

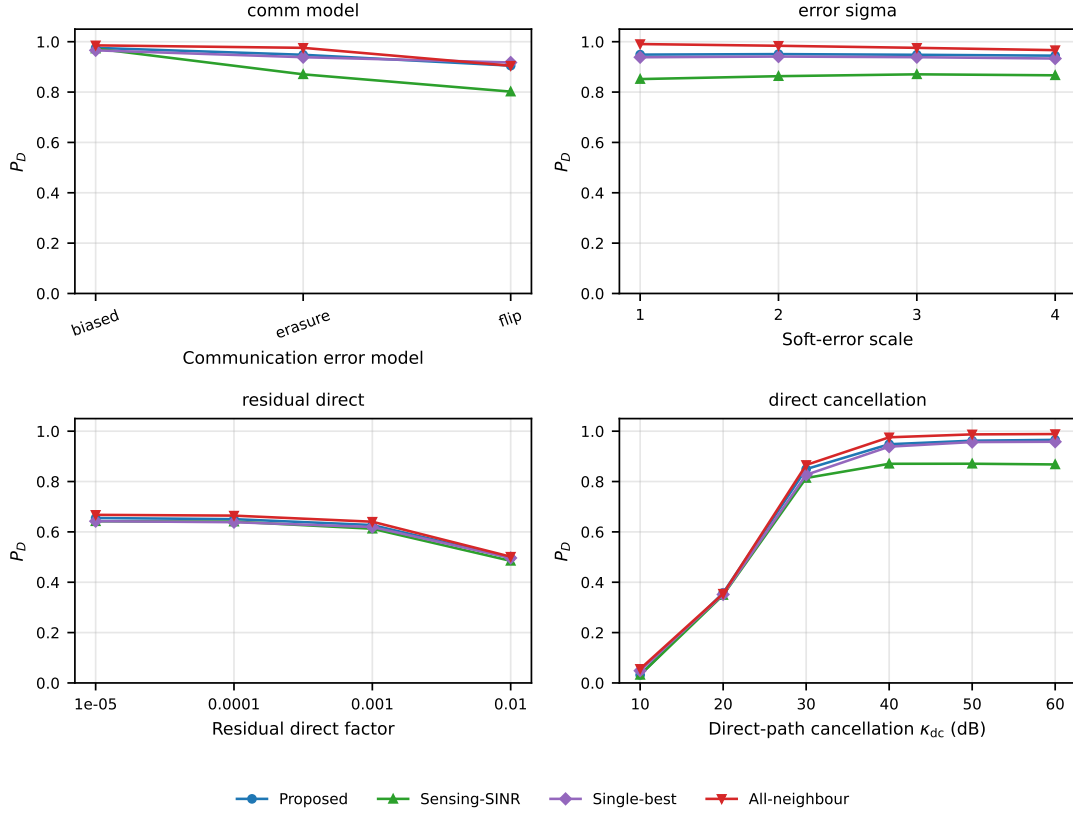


Fig. 5. Robustness along four axes: communication error model, soft-error scale, residual direct-interference floor (decoupled model) and direct-path cancellation (coupled model).

tonicity violations and zero submodularity violations, i.e. the objective (32) is empirically monotone submodular. Its total curvature is $c = 0.896$, giving the guarantee (33) $\frac{1-e^{-c}}{c} = 0.660$. On the small-scale instances where an exact search is tractable, the greedy rule achieves a mean 2.0% gap (median 0.2%, max 25.1%) from the same-objective optimum, confirming that the greedy selector is near-optimal on the very objective it optimises.

D. Interference Coupling and the Sensing-Suppression Budget

Since a UAV radiates one joint waveform into a shared band, the communication and the sensing receiver at the same node face the same concurrent transmitters over the same channel gains (5). At the main-comparison operating point the near-far ratio (6) is $\rho_{NF} = 41.3$ dB, the communication interference-to-noise ratio is +29.4 dB, and the direct-path contribution to the sensing interference under the default $\kappa_{dc} = 40$ dB is -3.9 dB. With the guard term corrected, coupling the two receivers raises the sensing interference by only 0.287 dB relative to the decoupled residual floors (-2.003 vs. -1.716 dB), so the structural correction to the sensing side is small. The coupling is nevertheless not free: it also restores the always-radiated sensing leakage at the communication receivers, and the illuminator's own direct path at the sensing receiver, which is why the corrected model is the more conservative of the two ($P_D = 0.9481$ against 0.9629 for the guard fix alone).

A second, unintended effect turned out to matter more. The SINR guard term, not the interference model, was dominating the sensing link budget: under the coupled model with the legacy guard the median sensing SINR is -22.0 dB, and moving the guard below the noise floor raises it to -10.1 dB and P_D from 0.6447 to 0.9481. Because the same guard entered the denominator of the interference-to-noise ratio, it had also deflated that ratio from -1.7 dB to -16.3 dB, which is why the sensing interference term was invisible in every earlier experiment.

Finally, κ_{dc} cannot be chosen freely, and Table III locates the resulting budget. The quantity that fixes it is the interference-to-noise ratio at the sensing receiver, $I_j^{s,dir}/N_0 = +36.1$ dB: the sensing task is viable once the direct path is cancelled down to the noise floor, and every additional 10 dB of suppression above that point buys nothing. Keeping everything else fixed, P_D rises from 0.0021 at 0 dB to 0.8494 at 30 dB and then saturates near 0.96 from 40 dB onwards. Even all-neighbour fusion collapses to $P_D = 0.0465$ at zero cancellation: no amount of link selection substitutes for direct-path suppression. Cooperative sensing therefore requires suppression of the order of ρ_{NF} , which is the price that sharing the spectrum with the communication function imposes on the sensing receiver. Reporting this threshold explicitly, instead of burying it in an unstructured interference floor, is what makes

TABLE III

DIRECT-PATH SUPPRESSION BUDGET OF THE SENSING RECEIVER, WITH ALL OTHER SETTINGS AT THE MAIN-COMPARISON OPERATING POINT.

κ_{dc} (dB)	0	10	20	30	40	80
$I_j^{s,dir}/N_0$ (dB)	+36.1	+26.1	+16.1	+6.1	-3.9	-43.9
P_D (proposed)	0.002	0.032	0.358	0.849	0.948	0.966
P_D (all-neighbour)	0.047	0.056	0.355	0.866	0.976	0.989

the scheduling problem an ISAC problem: the illumination that the communication function requires is the same illumination that limits the sensing function. Because the absolute P_D at a given κ_{dc} depends on the soft-statistic model — the strictly derived LLR model sits 4–6 points lower at the same operating point — the suppression budget, not the detection probability, is the model-robust statement.

The remaining figures and tables in this section use the decoupled interference floors of the earlier abstraction, for continuity with the frozen reference results. The coupled instantiation quantified above is the default in the released code and is available as a configuration preset; it shifts the absolute detection probabilities while preserving the method ordering and the qualitative conclusions. In particular, the proposed selector attains $P_D = 0.9481$ against 0.8705 for sensing-SINR selection and 0.9470 for Top-K deflection under the coupled model, versus 0.6506, 0.6408 and 0.6593 respectively under the decoupled one, while reducing the reporting delay from 688 ms to 6.9 ms.

V. CONCLUSION

This paper investigated communication-constrained cooperative sensing in a multi-UAV OTFS-ISAC network. We developed a communication-error-calibrated and C2F DD-aware Lagrangian link-selection method that jointly accounts for marginal deflection gain, reporting reliability, exchange delay, and DD-domain energy-capture loss. By calibrating exchanged soft statistics according to reporting reliability, the proposed method reduces the impact of unreliable reports on the fused statistic. The C2F DD-aware design further applies local DD-grid refinement only to shortlisted candidates, thereby recovering useful DD-domain energy while avoiding unnecessary fine-grid evaluations.

Simulation results showed that the proposed method retains most of the all-neighbor detection gain with much lower signaling overhead and exchange delay. Under the same per-target selected-link budget, it achieves higher detection probability than Sensing-SINR-based selection and provides detection performance comparable to the Top-K Deflection baseline with lower delay. Ablation and robustness results further support the benefits of communication-error calibration, target-priority weighting, delay-aware Lagrangian pricing, and C2F DD refinement.

Future work will extend the current geometry-driven soft-information abstraction toward waveform-level OTFS-ISAC receiver modeling, more realistic inter-UAV reporting proto-

cols, and joint detection-tracking design for dynamic multi-target UAV sensing networks.

REFERENCES

- [1] F. Liu *et al.*, “Integrated sensing and communications: Toward dual-functional wireless networks for 6G and beyond,” *IEEE J. Sel. Areas Commun.*, vol. 40, no. 6, pp. 1728–1767, Jun. 2022, doi: 10.1109/JSAC.2022.3156632.
- [2] N. Gonzalez-Prelcic *et al.*, “The integrated sensing and communication revolution for 6G: Vision, techniques, and applications,” *Proc. IEEE*, vol. 112, no. 7, pp. 676–723, Jul. 2024, doi: 10.1109/JPROC.2024.3397609.
- [3] K. Meng, C. Masouros, A. P. Petropulu, and L. Hanzo, “Cooperative ISAC networks: Performance analysis, scaling laws, and optimization,” *IEEE Trans. Wireless Commun.*, vol. 24, no. 2, pp. 877–892, Feb. 2025, doi: 10.1109/TWC.2024.3491356.
- [4] K. Meng, C. Masouros, G. Chen, and F. Liu, “Network-level integrated sensing and communication: Interference management and BS coordination using stochastic geometry,” *IEEE Trans. Wireless Commun.*, vol. 23, no. 12, pp. 19365–19381, Dec. 2024, doi: 10.1109/TWC.2024.3483031.
- [5] J. Mu, R. Zhang, Y. Cui, N. Gao, and X. Jing, “UAV meets integrated sensing and communication: Challenges and future directions,” *IEEE Commun. Mag.*, vol. 61, no. 5, pp. 62–67, May 2023, doi: 10.1109/MCOM.008.2200510.
- [6] K. Meng *et al.*, “UAV-enabled integrated sensing and communication: Opportunities and challenges,” *IEEE Wireless Commun.*, vol. 31, no. 2, pp. 97–104, Apr. 2024, doi: 10.1109/MWC.131.2200442.
- [7] A. Chawla, A. Patel, A. K. Jagannatham, and P. K. Varshney, “Distributed detection in massive MIMO wireless sensor networks under perfect and imperfect CSI,” *IEEE Trans. Signal Process.*, vol. 67, no. 15, pp. 4055–4068, Aug. 2019, doi: 10.1109/TSP.2019.2924588.
- [8] A. Chawla, A. S. Sarode, A. K. Jagannatham, and L. Hanzo, “Distributed parameter detection in massive MIMO wireless sensor networks relying on imperfect CSI,” *IEEE Trans. Wireless Commun.*, vol. 20, no. 1, pp. 506–519, Jan. 2021, doi: 10.1109/TWC.2020.3025877.
- [9] M. Chen, M.-M. Zhao, A. Liu, M. Li, and Q. Shi, “Joint node selection and resource allocation optimization for cooperative sensing with a shared wireless backhaul,” *IEEE Trans. Signal Process.*, vol. 73, pp. 67–82, 2025, doi: 10.1109/TSP.2024.3516709.
- [10] W. Yuan *et al.*, “From OTFS to DD-ISAC: Integrating sensing and communications in the delay-Doppler domain,” *IEEE Wireless Commun.*, vol. 31, no. 6, pp. 152–160, Dec. 2024, doi: 10.1109/MWC.018.2300607.
- [11] M. F. Keskin *et al.*, “Integrated sensing and communications with MIMO-OTFS: ISI/ICI exploitation and delay-Doppler multiplexing,” *IEEE Trans. Wireless Commun.*, vol. 23, no. 8, pp. 10229–10246, Aug. 2024, doi: 10.1109/TWC.2024.3370501.